

Section 10-1: Automation Workflow

• Automation

Refers to using software-based solutions to perform processes with minimal human intervention.

- In the cloud, automation can be used for the following processes:
 - Creating, cloning, or shutting down a VM instance or server cluster
 - Configuring a virtual network
 - Deploying a large and complex cloud environment

Why Automate? (1 of 2) ●

- Automation in the cloud helps achieve the following goals:

- **Faster deployment, adjustments, and corrections**
- **Better control**
- **Lower costs**
- **More secure**

- Modern cloud automation tools

Provide flexible and powerful features that help organizations improve performance and reduce

Why Automate? (2 of 2) ●

- Automation tools can perform the following maintenance and security tasks:

- Create snapshots and backups.
- Clone, resize, or remove resources.
- Apply patches and updates..
- Generate internal API communications.
- Restart or shut down VM instances.
- Enable or disable alerts.
- Detect signs of an attack.
- Quarantine compromised resources.

Infrastructure as Code (IaC) (1 of 2) ●

- A **runbook** is a set of documents that describe exact steps to complete procedures in an IT environment. ,It is **written for humans**

- A runbook is usually written to be read and followed by humans.

- The automated use of the runbook concept is called **IaC (infrastructure as code)**

In IaC, cloud configuration steps are written as code and deployed at the same time.

- IaC uses a **playbook for execute** the tasks-jobs-deployments,

- It is used in **automation**,It is **machine-readable**

A **playbook** is a collection of tasks executed in a specific order.

- **Orchestration sequencing**

Is an important part of designing effective IaC templates

Infrastructure as Code (IaC) (2 of 2) ●

- Typical environment phases can be deployed from the same **IaC script**:

- **Development** : Used for building and testing new features. //
- **Staging** : Used to test changes before going live. //
- **Production** : The live environment used by real users. //
- **Duplicate production** : A copy of production used for testing or recovery. //

- **IaC provides documentation**

Each automated configuration is recorded, so teams can review past changes and understand when and why a problem happened.

Automation Tools (1 of 5) ●

- Automation tools can be **built into the cloud platform** or provided by **third-party tools**.
- There are **two main categories of automation tools**:
 - **Provisioning** : Used to create resources automatically; **auto-scaling** is a common provisioning task.
 - **Configuration management** :Used to manage and keep systems configured correctly after deployment.
- **Popular automation tools** include:
 - AWS's CloudFormation
 - Azure's Resource Manager
 - GCP's Deployment Manager
 - Terraform

Automation Tools (3 of 5) ●

- The following tools are **commonly used for configuration management (CM)**:
 - Ansible
 - Chef
 - Puppet
 - SaltStack Config
 - AWS's Systems Manager
 - Azure's Automation
 - GCP's Config Controller

Automation Tools (4 of 5) ●

- **Differentiating factors** between automation tools include the following:

– Push model vs. pull model

In a **push model**, the server sends configuration changes to resources(nodes), while in a **pull model**, resources contact the server to get configuration details.

– Immutable vs. idempotent

Immutable systems are not changed after deployment, while **idempotent** systems can apply configurations multiple times and only make needed changes.

– Imperative vs. declarative

Imperative tools require exact step-by-step instructions, while **declarative** tools only describe the desired final state.

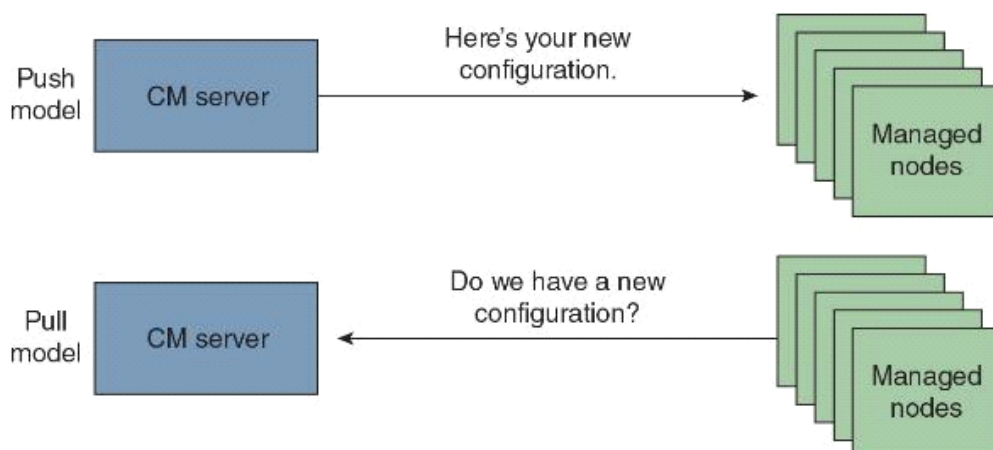


Figure 10-4 Push model versus pull model

Advanced Cloud Services (1 of 3) ●

– Data analytics

Is the study of raw data to find patterns, anomalies, and useful insights for business, research, and regulations.

– Cloud computing enables data analytics

Because the cloud provides almost unlimited processing power, storage, and network bandwidth.

– Big data

Very large data sets are called big data, and the three most common types are:

- **Social data** : Data collected from social media platforms.
- **Machine data** : Data collected from sensors, logs, and automated systems.
- **Transactional data** : Data related to purchases, payments, and invoices.

Advanced Cloud Services (2 of 3) ●

• Machine Learning (ML) and Artificial Intelligence (AI)

– AI (artificial intelligence)

Refers to a computer's ability to change its behavior based on ongoing input.

– ML (machine learning)

Is a process where computers learn from data instead of being explicitly programmed.

– **Main ML approaches** : There are three common learning approaches in ML:

- **Supervised learning** : The model learns from labeled data.
- **Unsupervised learning** : The model finds patterns in unlabeled data.
- **Reinforcement learning** : The model learns by trial and error using rewards and penalties.

Advanced Cloud Services (3 of 3) ●

• AI and ML in Automation

– Many automation tools in the cloud are used to **deploy AI and ML infrastructure**.

– **AI and ML support following automated tasks**, such as:

- **Application testing**
- **Resource scaling**
- **Network routing**
- **Security adaptations**

Section 10-2: Cloud Maintenance Processes ●

• **Common cloud maintenance tasks** include the following:

- **Resource monitoring** : Tracking cloud resources to ensure they work correctly and efficiently. //
- **Patch and update management** : Applying updates and fixes to keep systems secure and stable. //
- **Storage and backup management** : Managing data storage and ensuring backups are created and protected. //
- **Log management** : Collecting and reviewing logs to understand system activity and issues. //
- **Security monitoring and management** : Monitoring for security threats and managing access and protections. //
- **Troubleshooting** : Identifying and fixing problems in cloud services and resources. //

Cloud Maintenance Processes

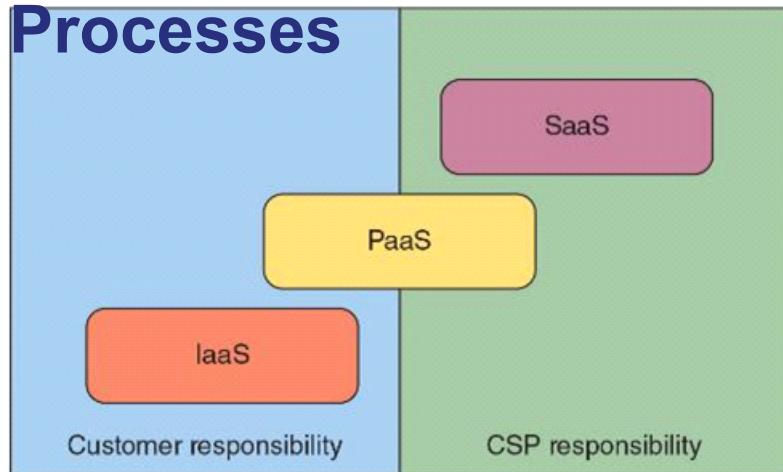


Figure 10-6 The distribution of maintenance responsibility varies by cloud service model

Lifecycle Management

- The following concepts are used to **manage resource life cycles**:
 - **Roadmaps** : It defines the major phases of a resource's life cycle.
 - **Version control** : A process used to track and manage changes over time.
 - **Updates and upgrades** : Regular updates fix bugs and close security gaps.
 - **Migrations** : As systems grow, they may need to move to a different platform.
 - **Deprecations** : Modular design helps plan and manage the removal of old features or system parts.

Types of Updates (1 of 2) ●

- Cloud elements that may require updates include the following:
 - VMs and server clusters
 - Networking components and virtual appliances
 - Storage components
 - Applications
 - Hypervisors and agents on VMs

Patching and Update Methodologies ●

- **Patch and update management** in the cloud follows standard procedures, including:
 - **Installing agents** : managing and controlling patching processes. //
 - **Scanning** : Systems are scanned to identify missing patches or vulnerabilities. //
 - **Testing** : Updates are tested. //
 - **Choosing update targets** : Specific systems or resources are selected for updates. //
 - **Considering backups** : Backups are checked or created. //
 - **Scheduling maintenance** : Updates are planned during low-impact time periods. //
 - **Monitoring compliance and completion rates** : Teams track the whole process. //

Automation and Security (1 of 3) ●

- Some companies **delay applying patches** until they are fully tested internally or tested in the market.
- Sometimes a **new patch fixes critical issues** and must be applied quickly.
- **Automating security** includes the following:
 - **Firewalls** : Default security groups can be set so new instances are protected automatically.
 - **IDS and IPS** : Cloud-native IDS and IPS tools are key security automation tools.
 - These tools can generate alerts and **automatically mitigate attacks**.

Automation and Security (2 of 3) ●

- **Automating security** continues with the following tools:
 - **FIM (File Integrity Monitoring)**
FIM creates a baseline of protected files and compares them over time to detect changes.
 - An alert is triggered if a difference is found.
 - **Log analytics**
Logs such as **VPC flow logs** capture security-related events and can trigger alerts.
 - **IAM scripting**
Scripts that interact with APIs or orchestrated services (like **AWS Lambda**) can automate user account management tasks.

Automation and Security (3 of 3) – Securing Automation ●

Securing automation focuses on keeping automation scripts safe and reducing security risks. Best practices for creating secure automation scripts include:

- **Accounts**
Each resource should have its own account or role with only the permissions it needs.
- **Passwords**
Passwords should not be written directly inside scripts.
- **Authentication**
Cloud platforms usually support key-based authentication, which is more secure than password-based authentication.

Automated Disaster Recovery ●

Automated disaster recovery helps keep cloud resources running during a crisis and restore full operation after a failure.

- The main goal is to **reduce problems during stressful situations** and prevent security threats or vulnerabilities from making the situation worse.
- **Availability** means how reliably and consistently authorized users can access a system or network resource.
- Availability is often measured by **uptime**, which shows how long a system stays online and working.
- Many **third-party providers** offer automated disaster recovery solutions in the cloud.
- **AWS CloudEndure** is an example of an automated disaster recovery service:
 - It can reduce **RPO (Recovery Point Objective)** to less than one second.
 - It can reduce **RTO (Recovery Time Objective)** to a few minutes.
 - It works by performing **automatic failover** when an outage happens.
Failover = yedek sisteme otomatik geçiş.